

LESSON 3-3 • SESSION 2 • PRACTICE • FORM A**The Bridge Strategy (Scaling Ratios)**

Learning target – I can use the bridge strategy to scale a ratio when I cannot multiply by a whole number. · 6.RP.A.3a

REMEMBER

- No whole number takes you to the target? **Build a bridge.**
- **Divide** both terms down to a smaller equivalent ratio, then **multiply** both terms up to the target.
- If needed, bridge all the way to **1**: 5 : 12 becomes 1 : 2.4.

WORKED EXAMPLE

A trail mix uses 6 cups of nuts for every 8 cups of raisins. How many cups of raisins go with 15 cups of nuts?

Step 1 $6 \times$ a whole number never equals 15.

Step 2 Bridge down: $6 : 8 \div 2 = 3 : 4$.

Step 3 Scale up: $3 : 4 \times 5 = 15 : 20$.

Answer **20 cups of raisins**

WORDS **bridge ratio** (*razón puente*) · **simplify** (*simplificar*) · **greatest common factor** (*máximo común divisor*)

PRACTICE

show your work · label units

1. Divide both terms by their GCF to write each bridge ratio.

a. $8 : 12 =$ _____

b. $21 : 28 =$ _____

c. $24 : 30 =$ _____

d. $16 : 36 =$ _____

2. **Orange Paint.** Orange paint uses 4 cups of red for every 6 cups of yellow. Use the bridge column to find the yellow for 10 cups of red.

Hint Divide by 2 to fill the bridge, then multiply by 5.

Red (cups)	4	2	10
Yellow (cups)	6		

Answer _____ cups yellow

3. **Salad Dressing.** A chef mixes 6 tbsp of oil with 9 tbsp of vinegar. The chef wants to use 14 tbsp of oil.

a. Bridge ratio: _____

b. Vinegar for 14 tbsp of oil: _____ tbsp

Show your work.

4. **Concrete Mix.** A worker mixes 9 bags of gravel with 15 bags of sand. How many bags of sand go with 21 bags of gravel?

Hint Divide both terms by 3 first.

Show your work.

Answer _____ bags of sand

5. **Candle Scents.** A candle uses 12 drops of lavender for 18 drops of vanilla. How many drops of vanilla go with 10 drops of lavender?

Hint Bridge to 2 : 3, then multiply by 5.

Show your work.

Answer _____ drops

6. **Mosaic Tiles.** A mural uses 24 blue tiles for every 30 cream tiles. How many cream tiles go with 20 blue tiles?

Show your work.

Answer _____ tiles

7. **Juice Machine.** A machine makes 12 liters of juice every 5 minutes.

Hint Bridge all the way to 1 minute first.

a. How many liters does it make in 1 minute? _____ L

b. How many liters does it make in 18 minutes? _____ L

Show your work.

8. **Smoothie Stand.** A stand uses 10 cups of fruit to make 4 smoothies.

a. How many cups of fruit does one smoothie use? _____ cups

b. How many cups are needed for 18 smoothies? _____ cups

c. Why can't you scale 10 : 4 by a whole number to reach 18 smoothies? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Aiden's Work.** A paint recipe uses 4 cups of red to 6 cups of yellow. To get 10 cups of red, Aiden added 6 to both terms: 10 : 12. Explain his mistake and find the correct amount of yellow.

TEST PRACTICE

choose the best answer

10. For the ratio 5 : 15, the amount that goes with 1 is –

A 5

B 15

C 10

D 3

11. The ratio of the first amount to the second is 4 : 10. If the first amount is 6, the second amount is –

A 12

B 15

C 24

D 60

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

21 B	35 R	16 Q	15 I	27 Z	25 D	17 X	45 S	43.2 E	2.4 G
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#3b	#4	#5	#6	#7a	#7b	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____